

Familiarisasi Hart Com 475



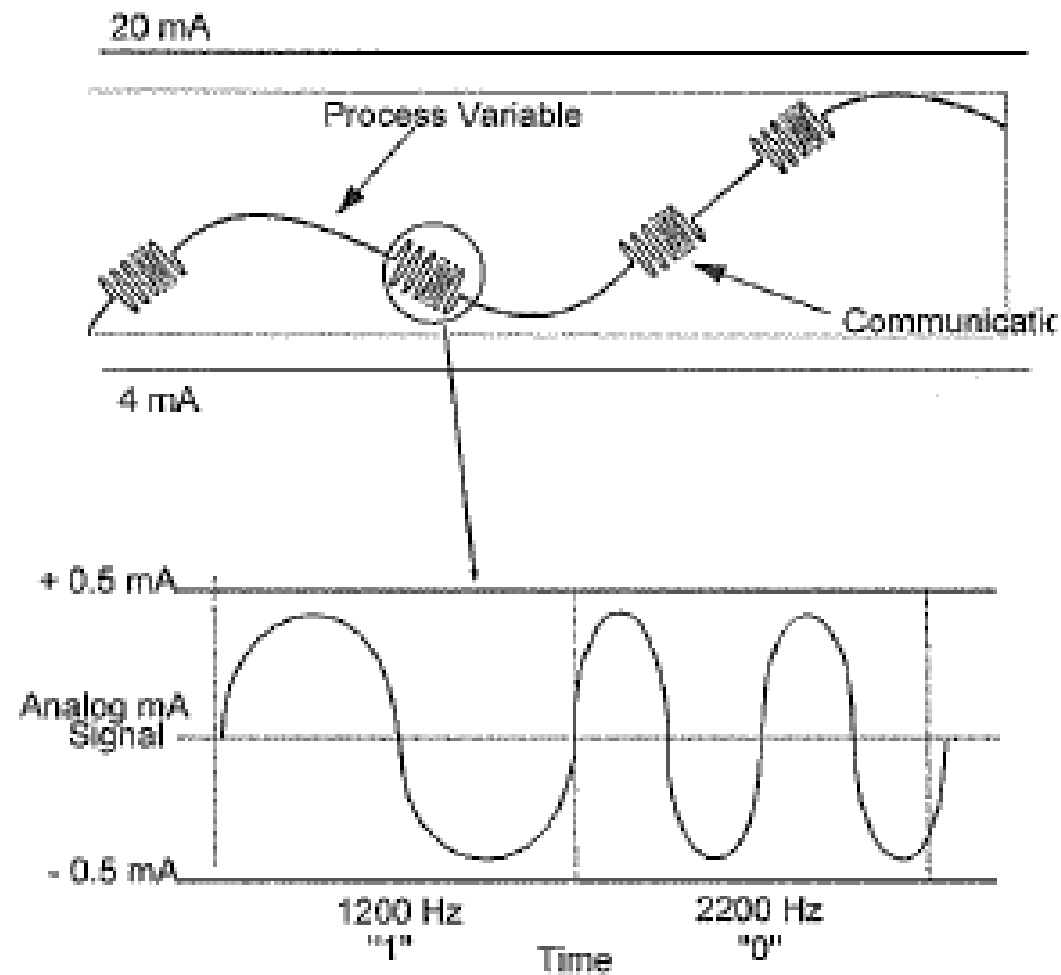
Apa itu HART?

HART merupakan singkatan dari “*Highway Addressable Remote Transducer*” yang bisa diterjemahkan sebagai Transducer berada pada remote area (jauh) . HART adalah sebuah protokol yang dikembangkan oleh *Rosemount* yang dibuat berdasarkan standar komunikasi *Bell 202*. HART merupakan open protokol yang berarti dapat digunakan dan dikembangkan oleh siapapun. HART memiliki dua modulasi sekaligus dalam komunikasinya yaitu menggunakan analog (4-20mA) dan Digital yaitu FSK (Frequency Shift Keying).

HART protocol

- Selain itu protokol HART juga menggunakan modulasi digital untuk berkomunikasi dengan cara mengirimkan data serial yang di modulasi dengan FSK (Frequency Shift Keying). data Analog di konversi kedalam digital kemudian dikirimkan melalui serial line dan dimodulasi. Sebelumnya, FSK adalah sebuah metode penyisipan data melalui gelombang sinyal dengan cara menyisipkan frekuensi. pada komunikasi HART, Frekuensi yang digunakan adalah 2200 Hz dan 1200Hz. frekuensi **2200Hz diterjemahkan sebagai '0'** dan **1200Hz diterjemahkan sebagai '1'**.

Hart FSK Signal Structure



Configure,Setting,Calibrate,ETC Smart Positioner with HART COMM 475

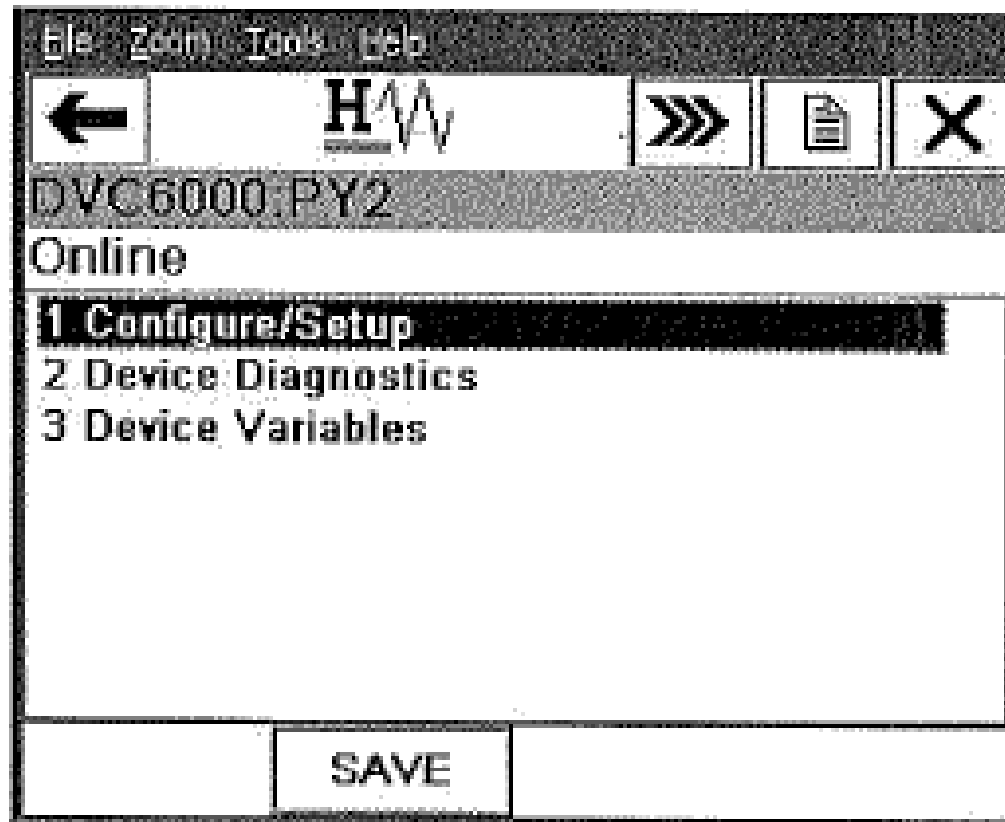




Viewing Device Information

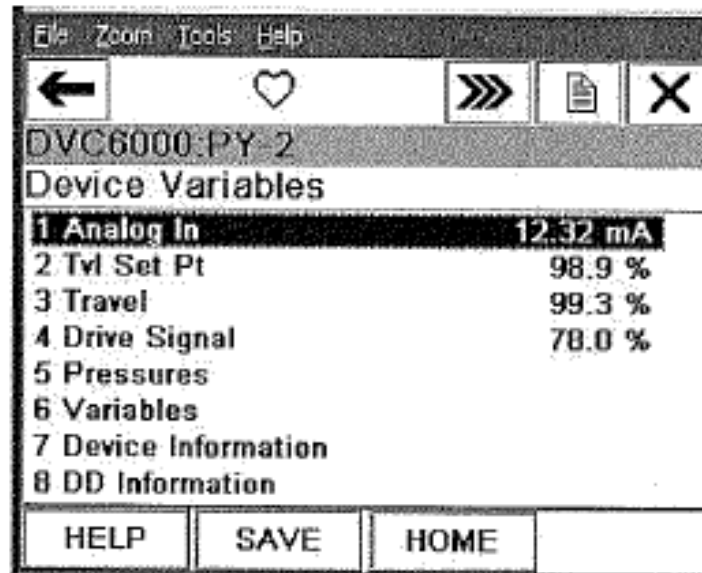
Using the handheld, you can display information that describes the DVC hardware and firmware revisions, and other important information relating vintage and features of the DVC and the DD in the 375 Field Communicator

Display and view device information



Display and view device information

Step 1. From the Online Menu, select (double-click) Device Variables. Note that this is a read-only screen.



DVC6000:PY-2	
Device Variables	
1 Analog In	12.32 mA
2 Tvl Set Pt	98.9 %
3 Travel	99.3 %
4 Drive Signal	78.0 %
5 Pressures	
6 Variables	
7 Device Information	
8 DD Information	

HELP SAVE HOME

This screen displays real time values for four important variables; analog input, travel setpoint, actual travel, and drive signal.

Display and view device information

Step 2. Double click on Device Information to show the following:



This window lists information that fully describes the DVC.



Detailed Setup

- ❖ Set Mode Parameter
- ❖ Activate and Deactivate Protection
- ❖ View Tuning Information
- ❖ View Travel Pressure Control Options
- ❖ Travel Limits and Cutoffs
- ❖ View and Manage Alerts
- ❖ Configure The Alert Record



Mode and Protection Menu

- Instrument Mode
- Control Mode
- Restart Control Mode
- Burst Mode

Activate and Deactivate Protection

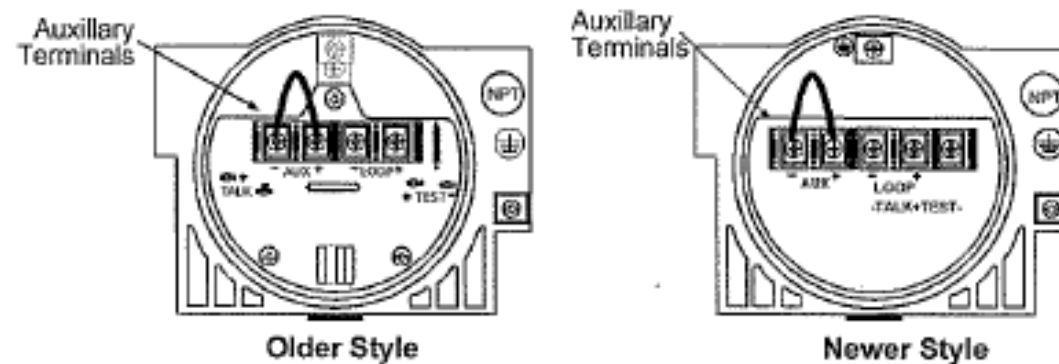


Figure 6-1. Configuration Protection Jumper Across AUX Terminals

Step 1. Activate Protection

- a) Press the Hot Key and select Protection, or from the Online Menu, select Configure/Setup → Detailed Setup → Mode and Protection → Protection.
- b) Observe the Warning and press OK.
- c) Select Config & Calib and press Enter.

Step 2. Observe the notice that protection has been sent to the instrument.

Step 3. Attempt to change the Valve Serial Number.

- a) From the Online Menu, select Configure/Setup → Detailed Setup Instrument → Valve & Actuator → Valve Serial Number



Response Control/Tuning Information

❖ Travel Tuning Set

Tvl Prop Gain—Travel Proportional Gain is the proportional gain for the travel control tuning set. Changing this parameter will also change the tuning set to Expert.

Tvl Velocity Gain—Travel Velocity Gain is the velocity gain for the travel control tuning set. Changing this parameter will also change the tuning set to Expert.

Tvl MLFB Gain—Travel MLFB Gain is the minor loop feedback gain for the travel control tuning set. Changing this parameter will also change the tuning set to Expert.

Response Control/Tuning Information

Tuning Set	Proportional Gain	Velocity Gain	Minor Loop Feedback Gain
C	4.4	3.0	35
D	4.8	3.0	35
E	5.5	3.0	35
F	6.2	3.1	35
G	7.2	3.6	34
H	8.4	4.2	31
I	9.7	4.85	27
J	11.3	5.65	23
K	13.1	6.0	18
L	15.5	6.0	12
M	18.0	6.0	12
X (Expert)	User Adjusted	User Adjusted	User Adjusted

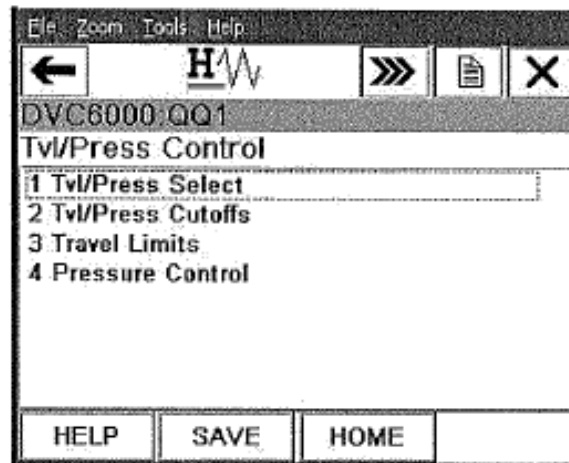
Response Control/Tuning Information

Actuator Manufacturer	Actuator Model	Actuator Size	Actuator Style	Starting Tuning Set	Travel Sensor Motion ⁽²⁾ Relay A or C ⁽³⁾	
Fisher	585C & 585CR	25 50, 60 60, 80 100, 130	Piston Dbl w/ or w/o Spring. See actuator instruction manual and nameplate.	F J L M	User Specified	
	657	30 34, 40 45, 50 46, 60, 70, 76, & 80-100	Spring & Diaphragm	H K L M	Away from the top of the instrument	
	667	30 34, 40 45, 50 46, 60, 70, 76, & 80-100	Spring & Diaphragm	H K L M	Towards the top of the instrument	
	1051 & 1052	20, 30 33 40 60, 70	Spring & Diaphragm (Window-mount)	H I K M	Away from the top of the instrument	
	1061	30 40 60 68, 80, 100, 130	Piston Dbl w/o Spring	J K L M	Depends upon pneumatic connections. See description for Travel Sensor Motion	
	1066	20, 27, 75	Piston Dbl w/o Spring	Specify	Depends upon pneumatic connections. See description for Travel Sensor Motion	
	1066SR	20 27, 75	Piston Sgl w/Spring	G L	Mounting Style	Travel Sensor Motion
					A	Away from the top of the instrument
					B	Towards the top of the instrument
					C	Towards the top of the instrument
					D	Away from the top of the instrument
	2052	1 2 3	Spring & Diaphragm (Window-mount)	H J M	Away from the top of the instrument	
	3024	GA 1.21 GA 1.31 GA 1.41	Spring & Diaphragm	E H K	For P ₀ operating mode (air opens): Towards the top of the instrument For P ₂ operating mode (air closes): Away from the top of the instrument	
GX	225	Spring & Diaphragm	X ⁽¹⁾	Air to Open	Air to Close	
	750		K	Towards the top of the instrument	Away from the top of the instrument	
	1200		M			
Baumann	Air to Extend	16 32	Spring & Diaphragm	C E H	Towards the top of the instrument	
	Air to Retract	54			Away from the top of the instrument	
	Rotary	10 25 54		E H J	Specify	

NOTE: Refer to figure 2-3 and table 4-8 for feedback connection (magnet assembly) information.
1. X = Expert Tuning, Proportional Gain = 4.2; Velocity Gain = 3.0; Minor Loop Feedback Gain = 18.0
2. Travel Sensor Motion in this instance refers to the motion of the magnet assembly.
3. Values shown are for Relay A and C. Reverse for Relay B.

Travel Pressure Control Options

Step 1. From the Response Control Menu, select Tvl/Press Control. The following window is displayed.



Step 2. Explore the following options.

Tvl/Press Select – This option lets you choose Travel Control, Pressure Control, Tvl/Press Auto Recv, and Tvl/Press Man Recv.

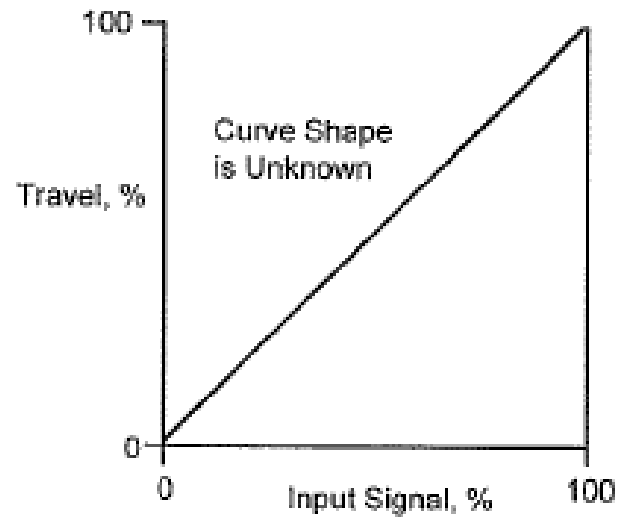
Tvl/Press Cutoffs – This choice lets you set the hi and low pressure cutoffs. These cutoffs are in affect for both travel and pressure control. Cutoffs are the upper and lower saturation points of the device.

Travel Limits - This choice lets you set the hi and low travel limits; applies only to Travel Control. Think of limits as soft stops.

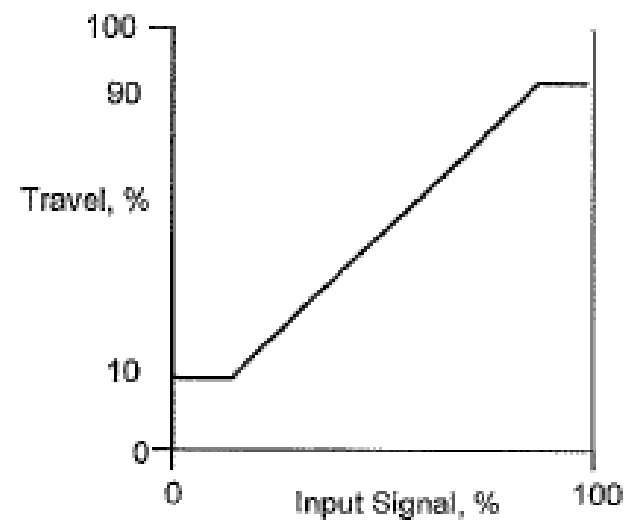
Pressure Control – Allows the user to view and edit the pressure range hi and pressure range lo values.

Travel Limits

- Travel Limit Low
- Travel Limit High



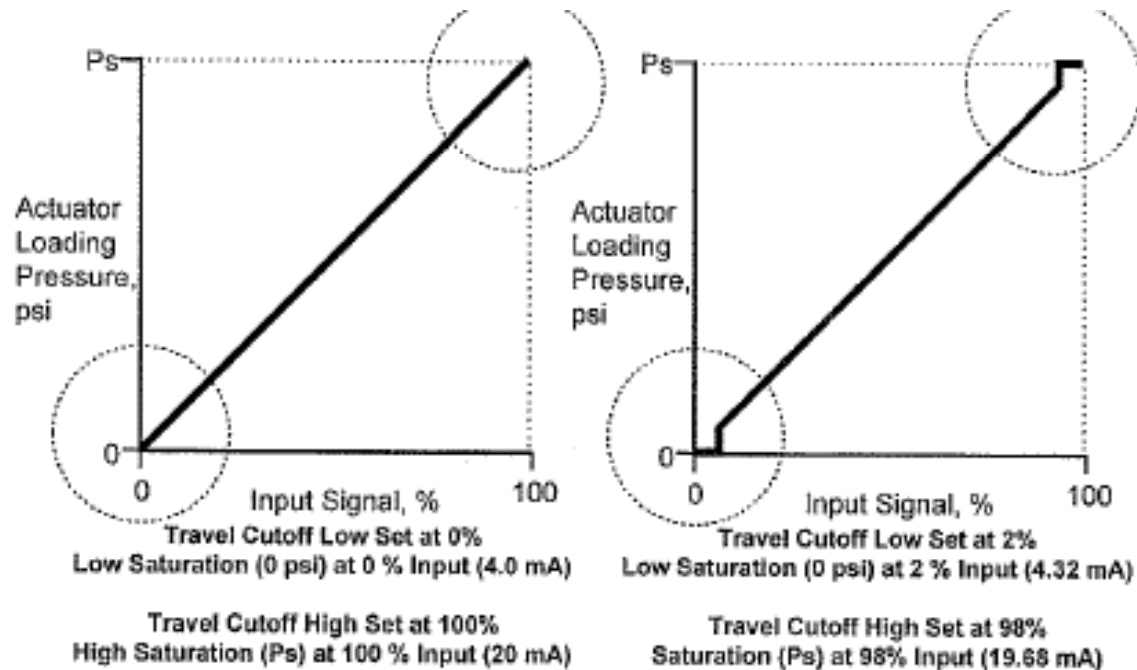
Travel Limits Not Enabled



Travel Limits of 10 and 90%

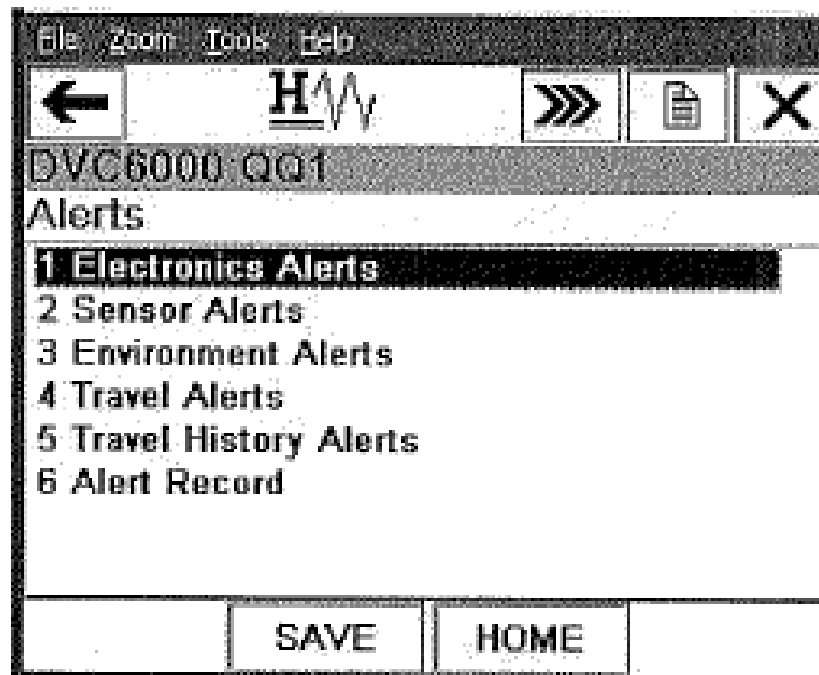
Travel Cutoffs

- Travel Cutoff low
- Travel Cutoff High



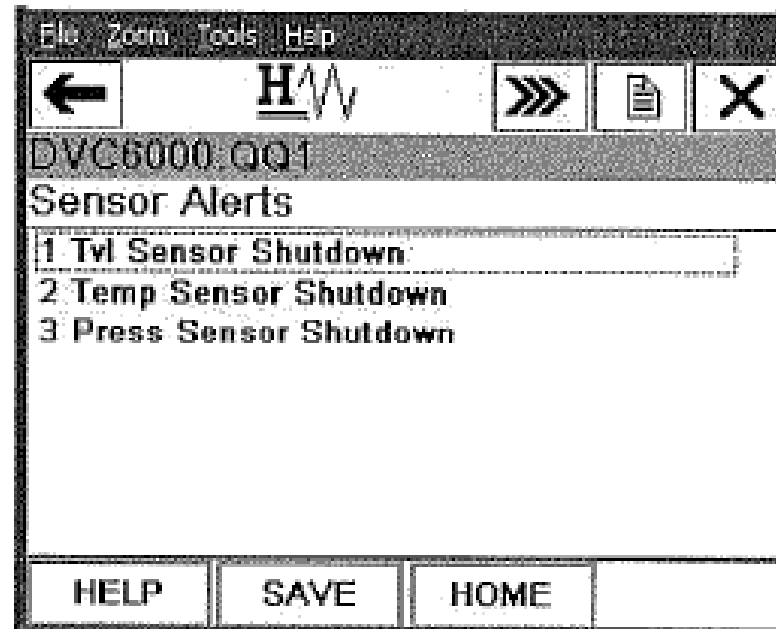
View and Manage Alerts

Step 1. To view the Alerts Menu, select the following: Online → Configure/Setup → Detailed Setup → Alerts



View and Manage Alerts

Step 1. Select Sensor Alerts and view the menu.

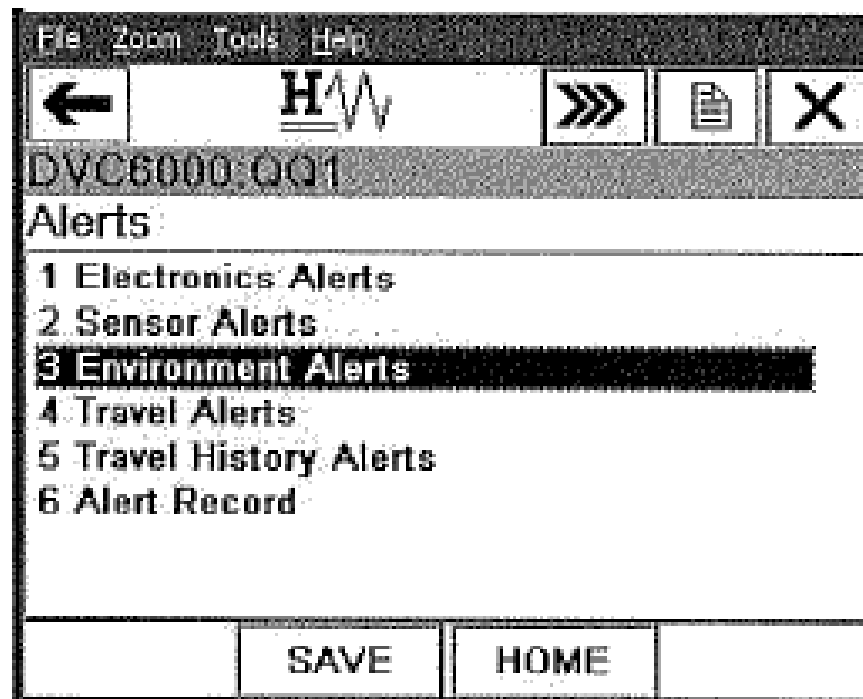


Notice that these are all Shutdown alerts; i.e., if they are enabled and problems with any of the sensors are detected, the DVC will shutdown.

Step 2. Click the back arrow to return to the Alerts menu.

View and Manage Alerts

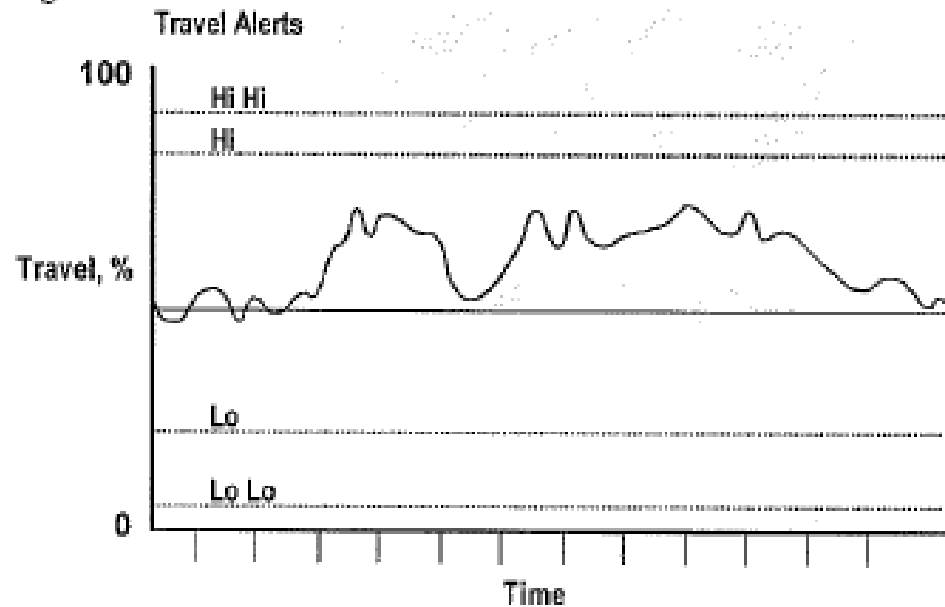
Step 1. Double-click on Environment Alerts



View and Manage Alerts

Travel Alerts

The DVC can be configured to alert four different travel points as shown in the figure below.





Calibration

- Auto Calibration
- Manual Calibration

Auto Calibration

Field Communicator

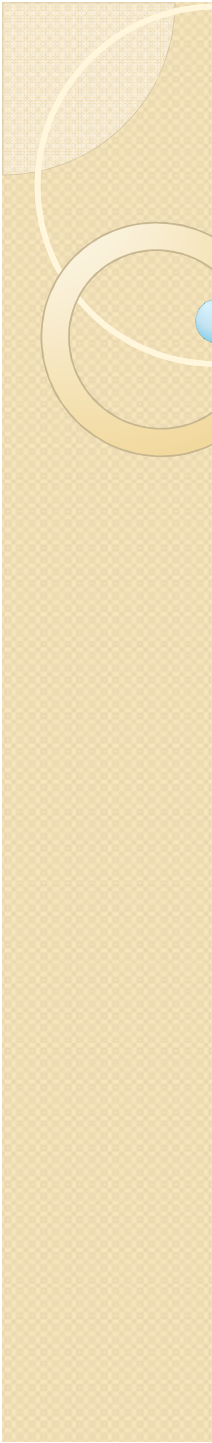
Configure / Setup > Calibrate > Travel Calibration > Auto Trl Calib (1-3-1-1)

1. The auto calibration procedure is automatic. It is completed when the *Calibrate* menu appears.

During calibration, the instrument seeks the high and low end points and the minor loop feedback (MLFB) a bias. By searching for the end points, the instrument establishes the limits of physical travel, i.e. the actual 100% positions. This also determines how far the relay beam swings to calibrate the sensitivity of the beam sensor.

2. Place the instrument In Service and verify that the travel properly tracks the current source.

If the unit does not calibrate, refer to table [5-1](#) for error messages and possible remedies.



Manual Calibration

- Analog Adjust
- Digital Adjust